THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The chord AB subtends 98° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 121° **(b)** 141° **(c)** 131° **(d)** 151°

2. In a cyclic quadrilateral $ABCD$, $\angle A = 68^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 78° **(b)** 58° **(c)** 88° **(d)** 68°

3. In triangle OAB, $OA = OB$ and $\angle AOB = 100^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 40° **(b)** 50° **(c)** 30° **(d)** 60°

4. A line from the centre of a circle that bisects a chord (not a diameter) is: 1

(a) parallel to the chord **(b)** a diameter always
(c) equal to the chord **(d)** perpendicular to the chord

5. A chord is at distance 15 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 39 **(b)** 41 **(c)** 40 **(d)** 42

6. The radius of a circle is 26 cm and a chord is 48 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 9 **(b)** 11 **(c)** 10 **(d)** 12

7. The chord AB subtends 112° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 76° **(b)** 56° **(c)** 66° **(d)** 46°

8. The chord AB subtends 116° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 122° **(b)** 132° **(c)** 112° **(d)** 142°

9. In a cyclic quadrilateral $ABCD$, $\angle A = 84^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 104° **(b)** 74° **(c)** 94° **(d)** 84°

10. In triangle OAB, $OA = OB$ and $\angle AOB = 120^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 50° **(b)** 20° **(c)** 30° **(d)** 40°

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A chord AB of a circle with centre O subtends an angle of 90° at O. Find the angle it subtends at a point C on the minor arc. **2**

12. $ABCD$ is a cyclic quadrilateral and BC is produced to E. If $\angle A = 74^\circ$, find $\angle DCE$. **2**

13. In a circle of radius 29 cm, a chord is 20 cm from the centre. Find its length. **2**

14. A chord AB of a circle with centre O subtends an angle of 100° at O. Find the angle it subtends at a point C on the minor arc. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In triangle OAB with O the centre of a circle, $\angle AOB = 110^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. **3**

16. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 20)^\circ$ and $\angle C = (3x - 10)^\circ$. Find x and $\angle A$. **3**

17. A chord of length 32 cm is 12 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. **3**

OR

In triangle OAB with O the centre of a circle, $\angle AOB = 130^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. **3**

18. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 5)^\circ$ and $\angle C = (3x - 5)^\circ$. Find x and $\angle A$. **3**

OR

A chord of length 24 cm is 16 cm from the centre of a circle. Find the radius and the area of the right triangle formed by the radius, the perpendicular from the centre and half the chord. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. Find the distance between them. **5**

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 82^\circ$, $\angle B = 71^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. **5**

20. Two parallel chords of lengths 30 cm and 40 cm lie on the same side of the centre of a circle of radius 25 cm. Find the distance between them. **5**

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 58^\circ$, $\angle B = 103^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. **5**

END OF QUESTION PAPER — Total: 40 Marks